Chapter 5

Biostatistics and Data Analysis

Contents

- Standard Deviation
- Standard Error
- Z score Outlier calculation

Standard Deviation

Standard Deviation

- The Standard Deviation is a measure of how spread-out numbers are.
- It is the **square root** of the **Variance.**

Steps for Finding Sample Variance & Sample Standard Deviation

1. Find the mean of the sample data set. $\bar{x} = \frac{\Sigma x}{n}$

2. Find deviation of each entry. $x - \bar{x}$

3. Square each deviation. $(x - \bar{x})^2$

4. Add to get the sum of the deviations squared. $\Sigma(x - \bar{x})^2$

5. Divide by $n - 1$ to get the **sample variance**. $s^2 = \frac{\Sigma(x - \bar{x})^2}{n - 1}$

6. Find the square root to get the **sample standard deviation**. $s = \sqrt{\frac{\Sigma(x - \bar{x})^2}{n - 1}}$

Steps for Finding Population Variance & Population Standard Deviation

1. Find the mean of the Population data set. $\mu = \frac{\sum X}{N}$

2. Find deviation of each entry. $(X - \mu)$

3. Square each deviation. $(X - \mu)^2$

4. Add to get the sum of the deviations squared. $\sum (X - \mu)^2$

5. Divide by N to get the **Population variance**. $\sigma^2 = \frac{\sum (X - \mu)^2}{N}$

6. Find the square root to get the **Population standard deviation**. $\sigma = \sqrt{\sigma^2}$

Sample Variance and Sample Standard Deviation

Sample Variance

$$s^2 = \frac{\Sigma(x - \bar{x})^2}{n - 1}$$

Sample Standard Deviation

$$s = \sqrt{s^2} = \sqrt{\frac{\Sigma(x - \bar{x})^2}{n - 1}}$$

Population Variance and Population Standard Deviation

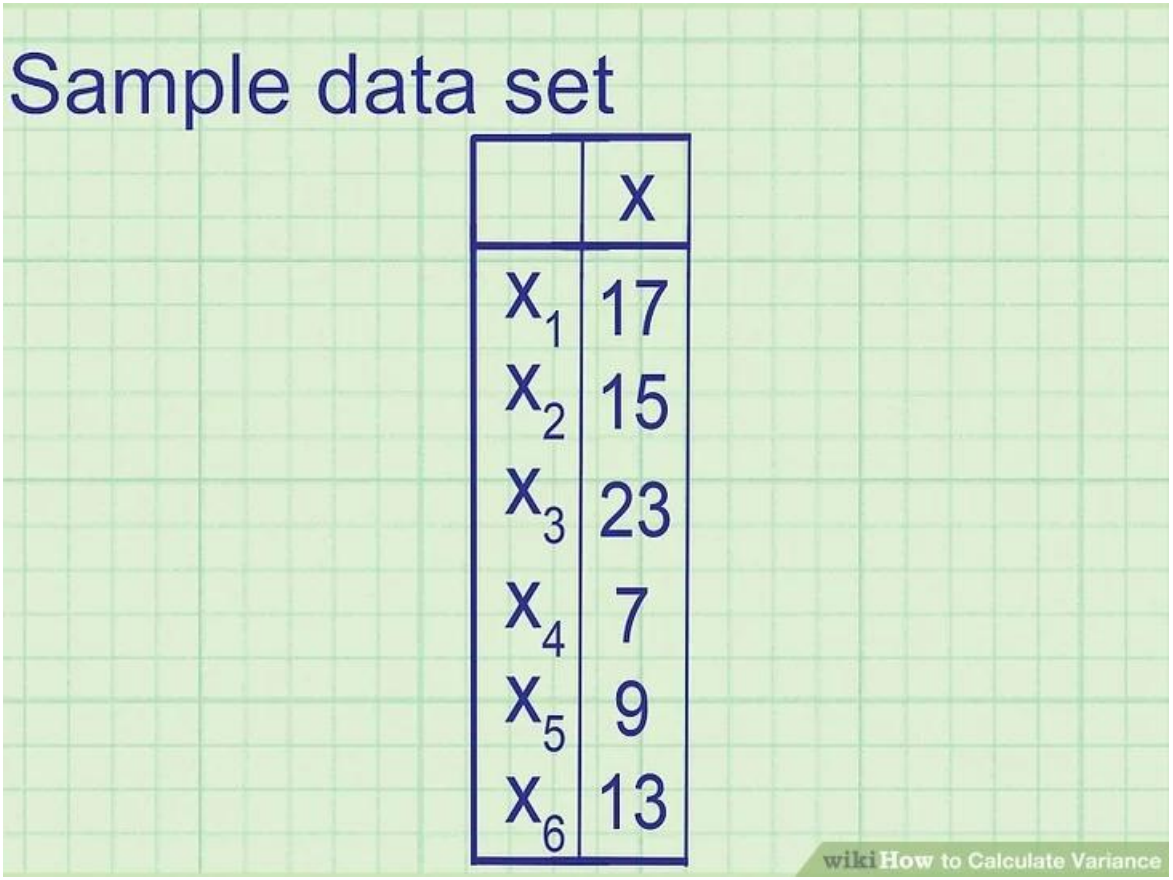
Population Variance

$$\sigma^2 = \frac{\Sigma(x-\mu)^2}{N}$$

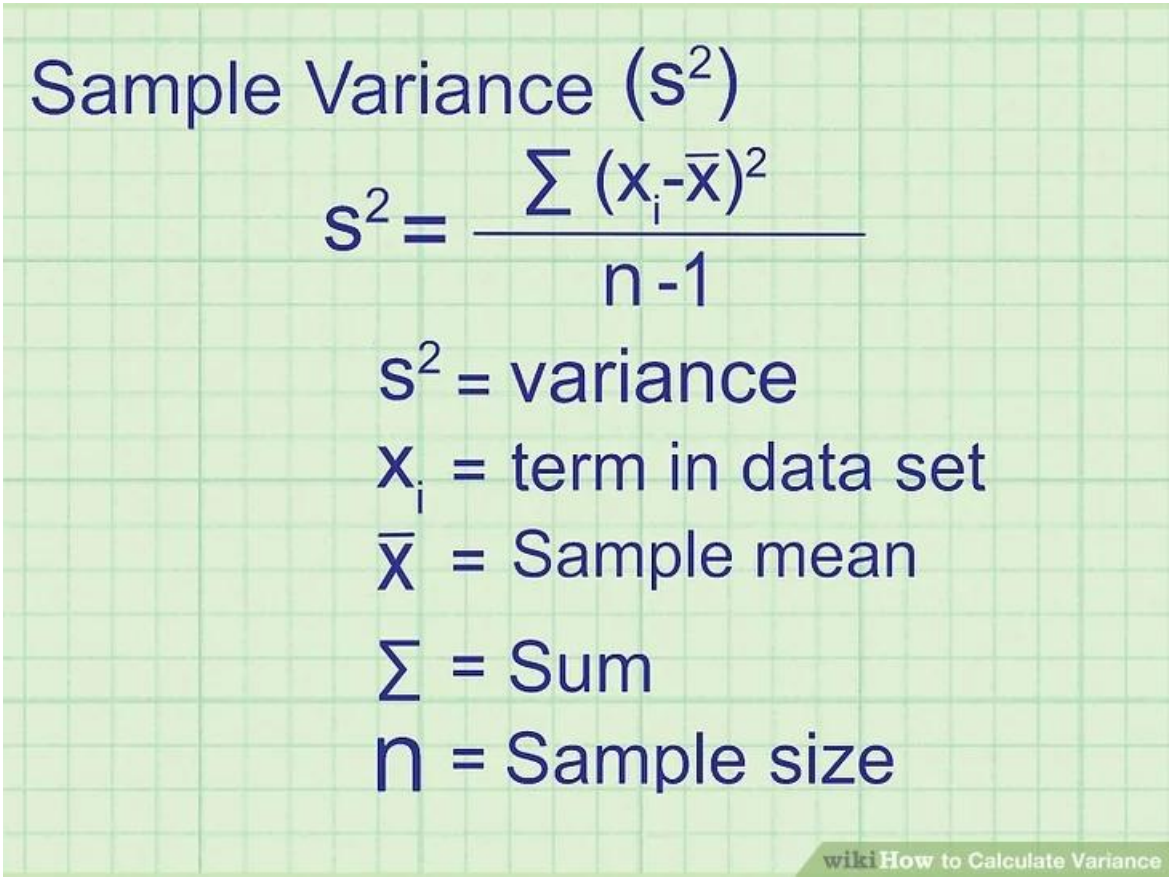
Population Standard Deviation

$$\sigma = \sqrt{\sigma^2} = \sqrt{\frac{\Sigma(x-\mu)^2}{N}}$$

Calculate Sample Variance

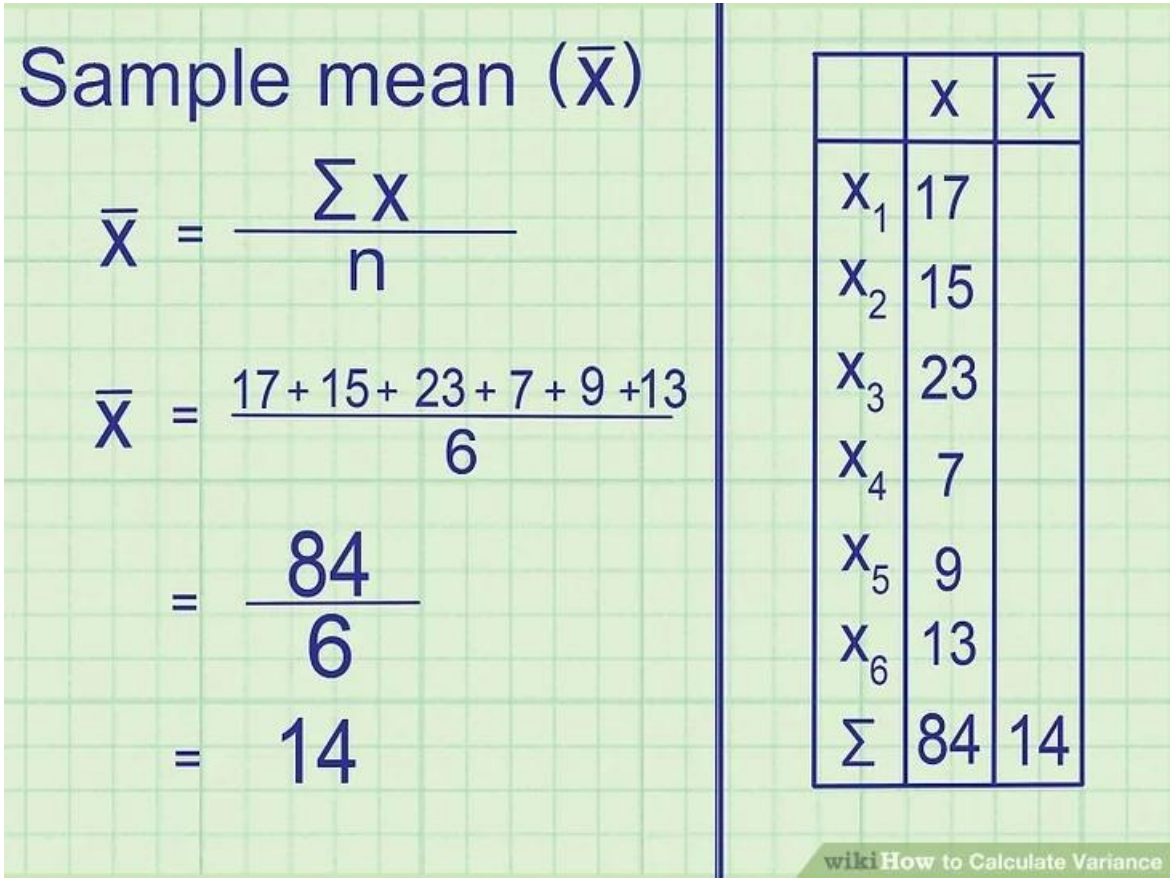


Calculate Sample Variance



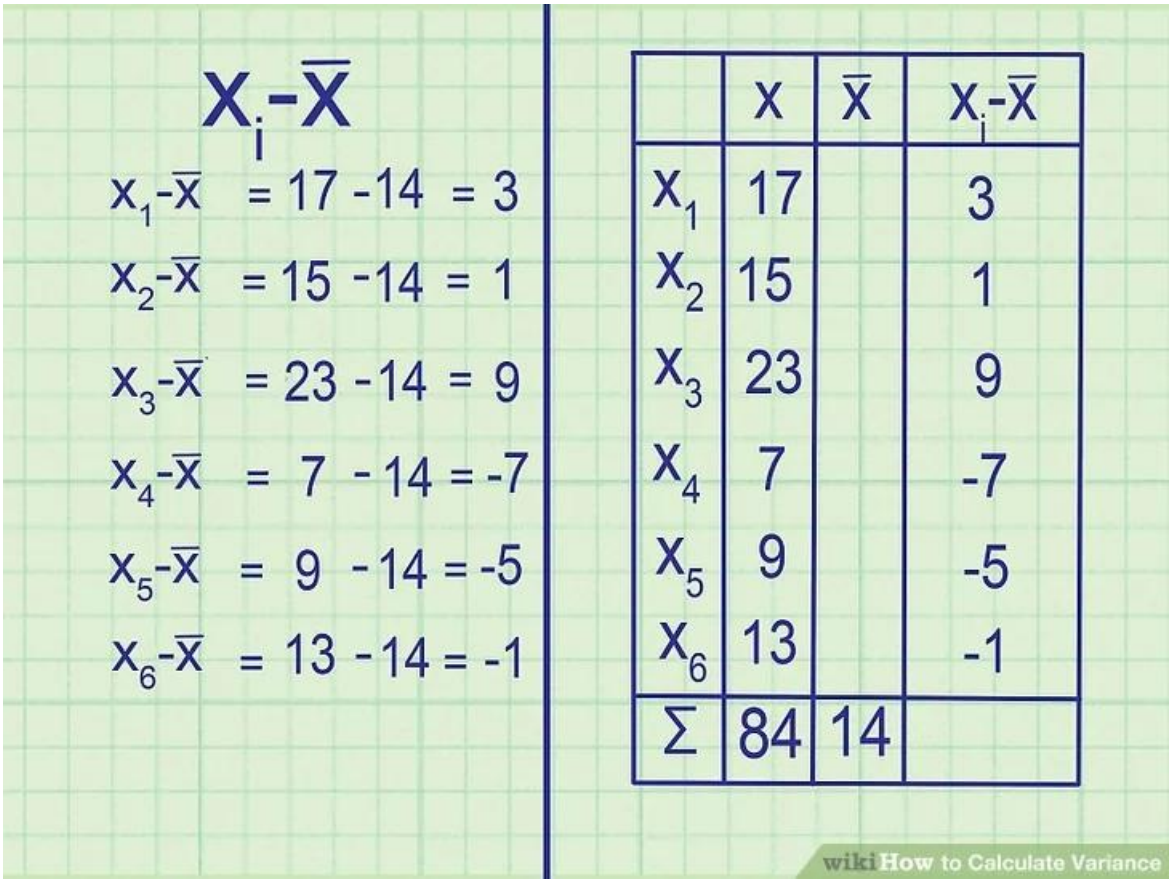
Calculate Sample Variance

1. Find the mean of the sample data set. $\bar{x} = \frac{\Sigma x}{n}$



Calculate Sample Variance

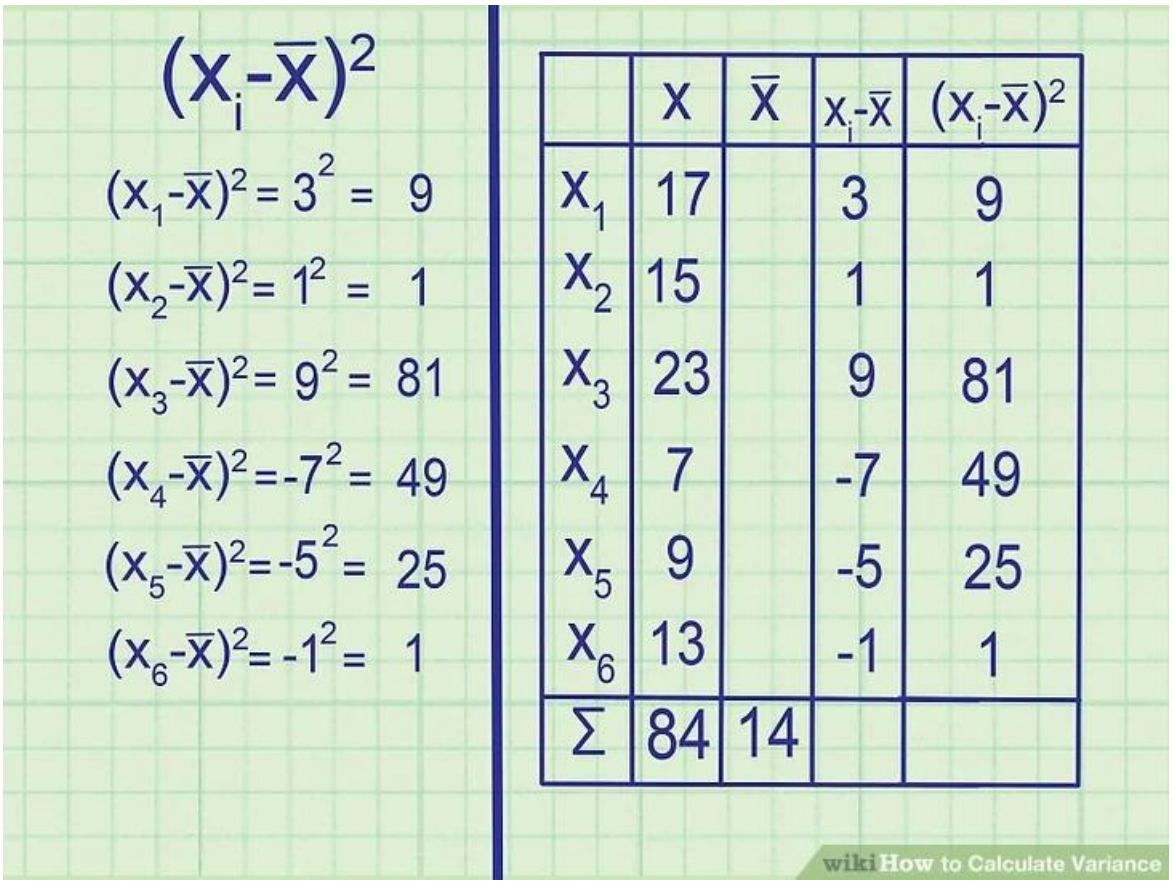
2. Find deviation of each entry. $x - \overline{x}$



	x	$\overline{x}$	$x_i - \overline{x}$
x_1	17		3
x_2	15		1
x_3	23		9
x_4	7		-7
x_5	9		-5
x_6	13		-1
Σ	84	14	

Calculate Sample Variance

3. Square each deviation. $(x-\bar{x})^2$



	x	$\bar{x}$	$x_i-\bar{x}$	$(x_i-\bar{x})^2$
x_1	17		3	9
x_2	15		1	1
x_3	23		9	81
x_4	7		-7	49
x_5	9		-5	25
x_6	13		-1	1
Σ	84	14		

Calculate Sample Variance

4. Add to get the sum of the deviations squared. $\Sigma(x-\bar{x})^2$

$$\sum (x_i-\bar{x})^2$$

$$= (x_1-\bar{x})^2 + (x_2-\bar{x})^2 + (x_3-\bar{x})^2 + (x_4-\bar{x})^2 + (x_5-\bar{x})^2 + (x_6-\bar{x})^2$$

$$= 9 + 1 + 81 + 49 + 25 + 1$$

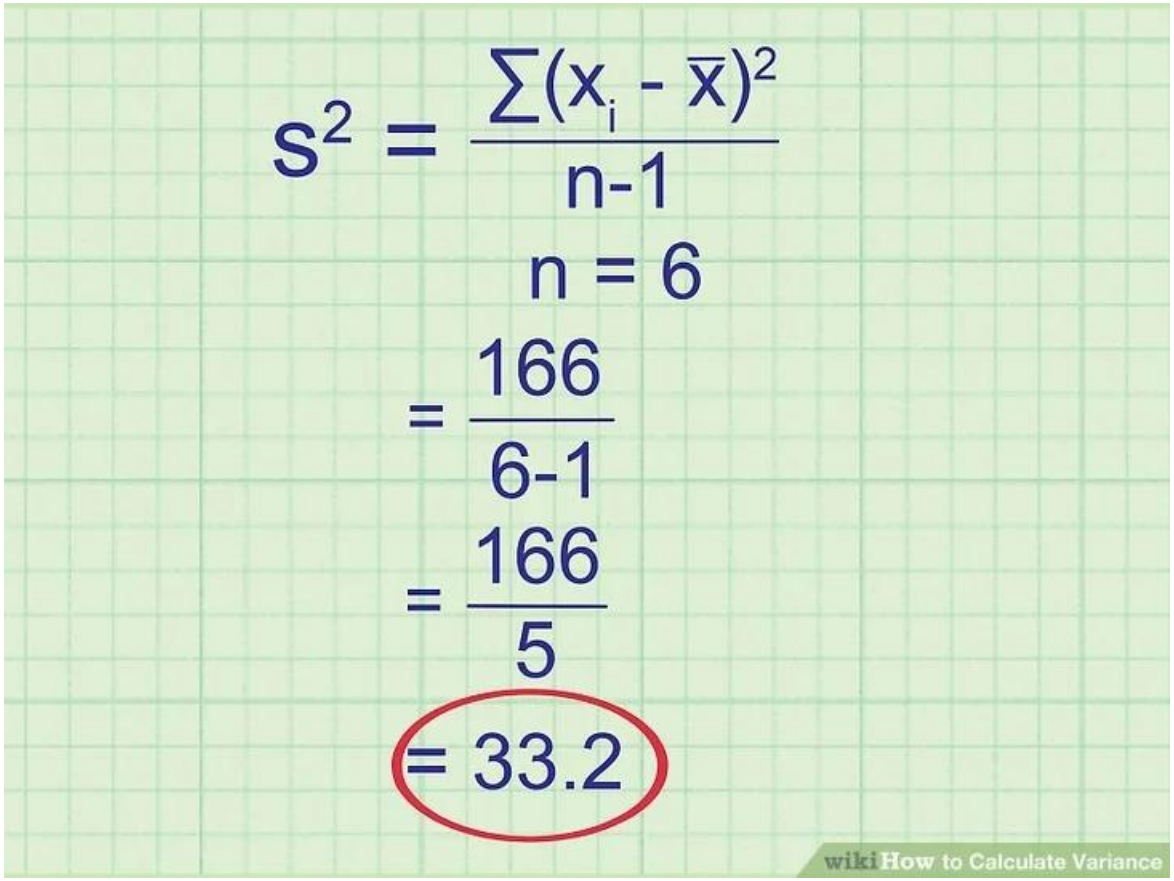
$$= 166$$

	x	$\bar{x}$	$x_i-\bar{x}$	$(x_i-\bar{x})^2$
x_1	17		3	9
x_2	15		1	1
x_3	23		9	81
x_4	7		-7	49
x_5	9		-5	25
x_6	13		-1	1
Σ	84	14		166

wikiHow to Calculate Variance

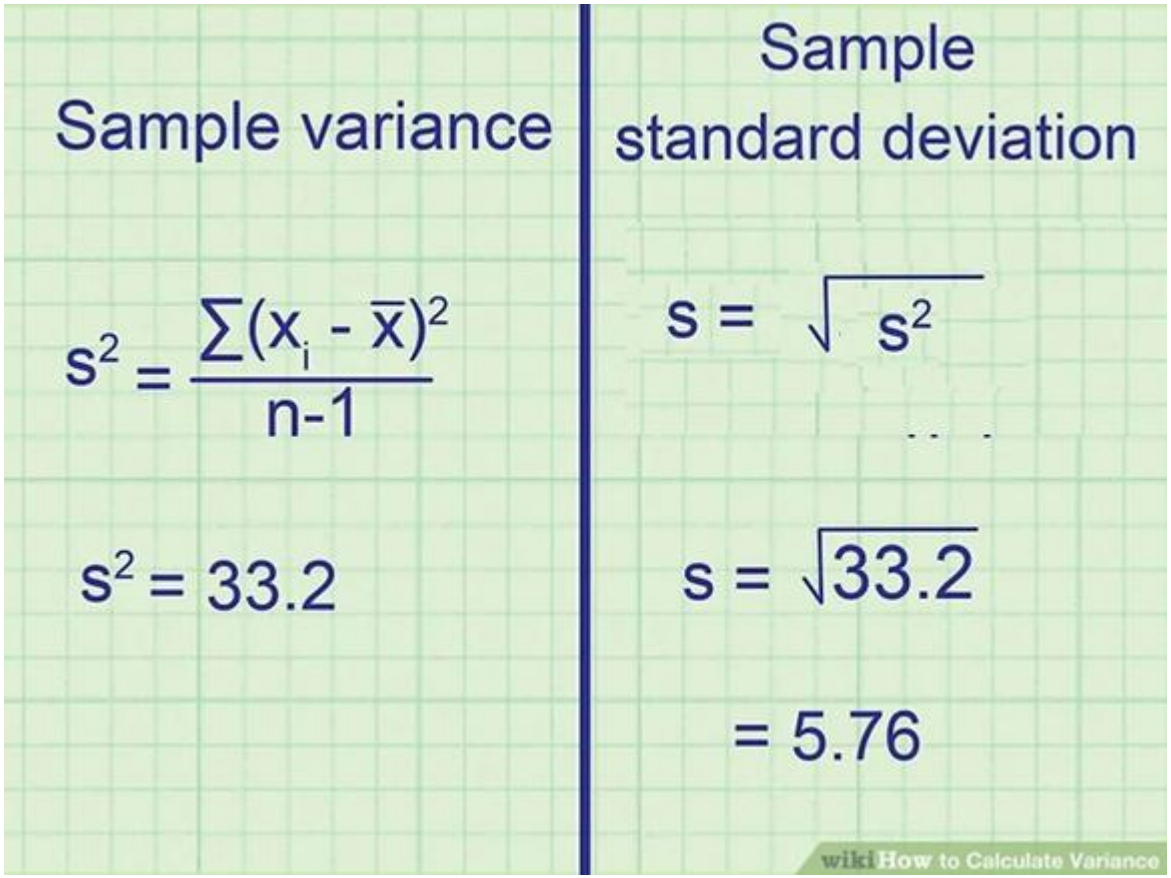
Calculate Sample Variance

5. Divide by $n - 1$ to get the **sample variance**. $s^2 = \frac{\Sigma(x - \bar{x})^2}{n-1}$



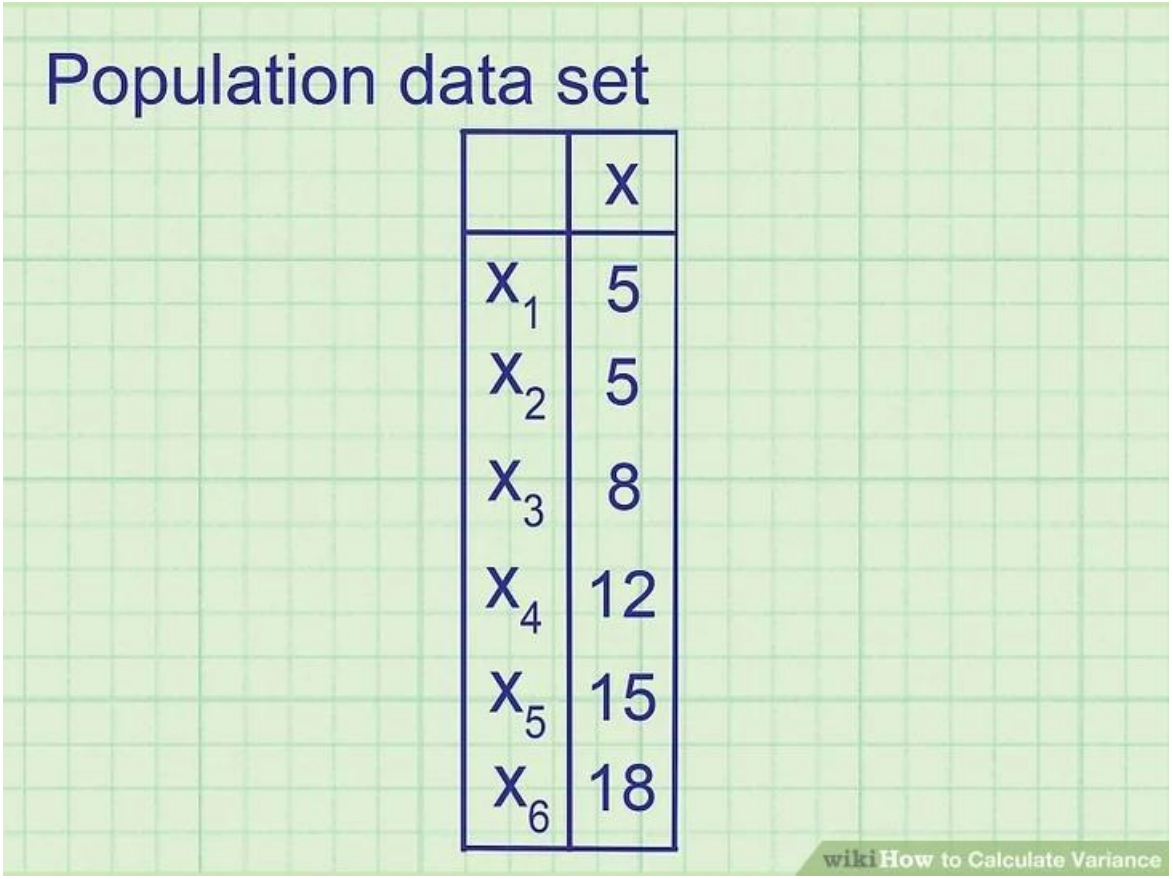
Calculate Sample Variance

6. Find the square root to get the **sample standard deviation**. $s = \sqrt{\frac{\Sigma(x-\bar{x})^2}{n-1}}$

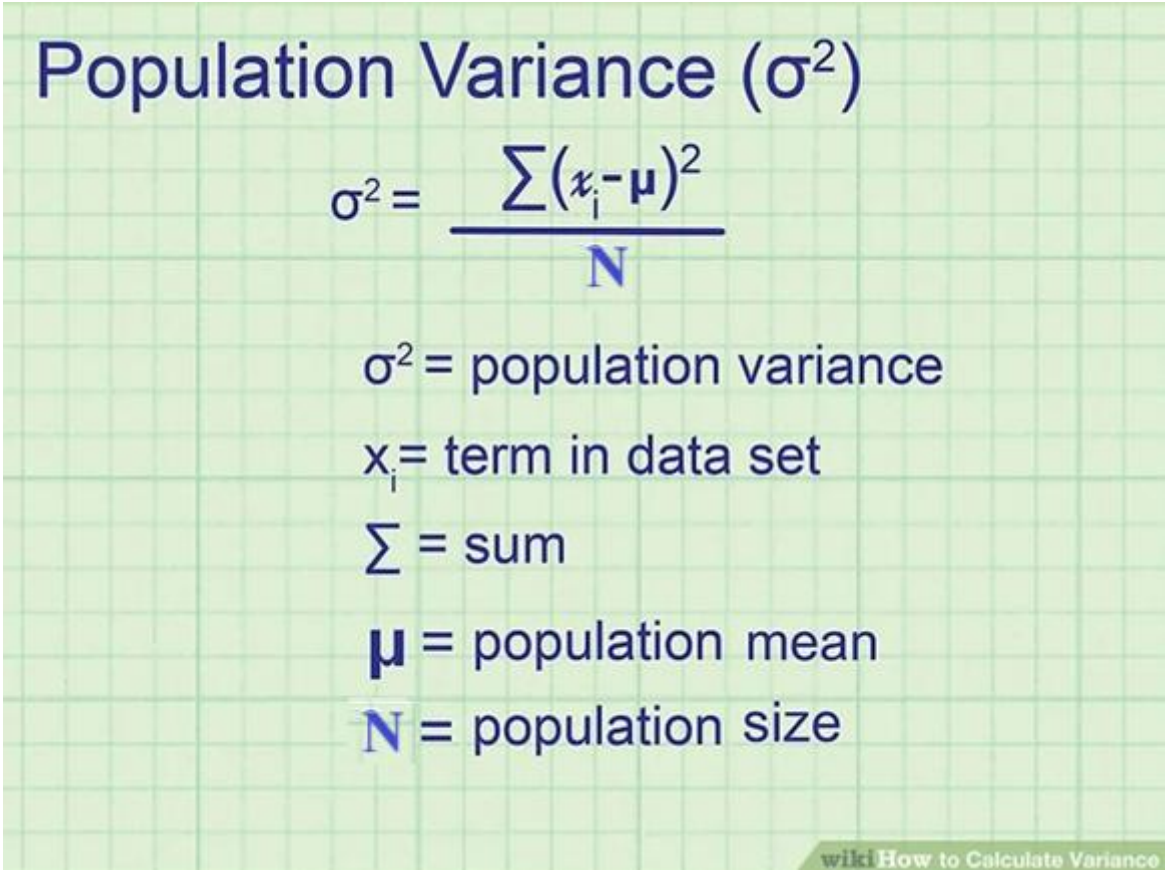


Calculate Population Variance

Calculate Population Variance

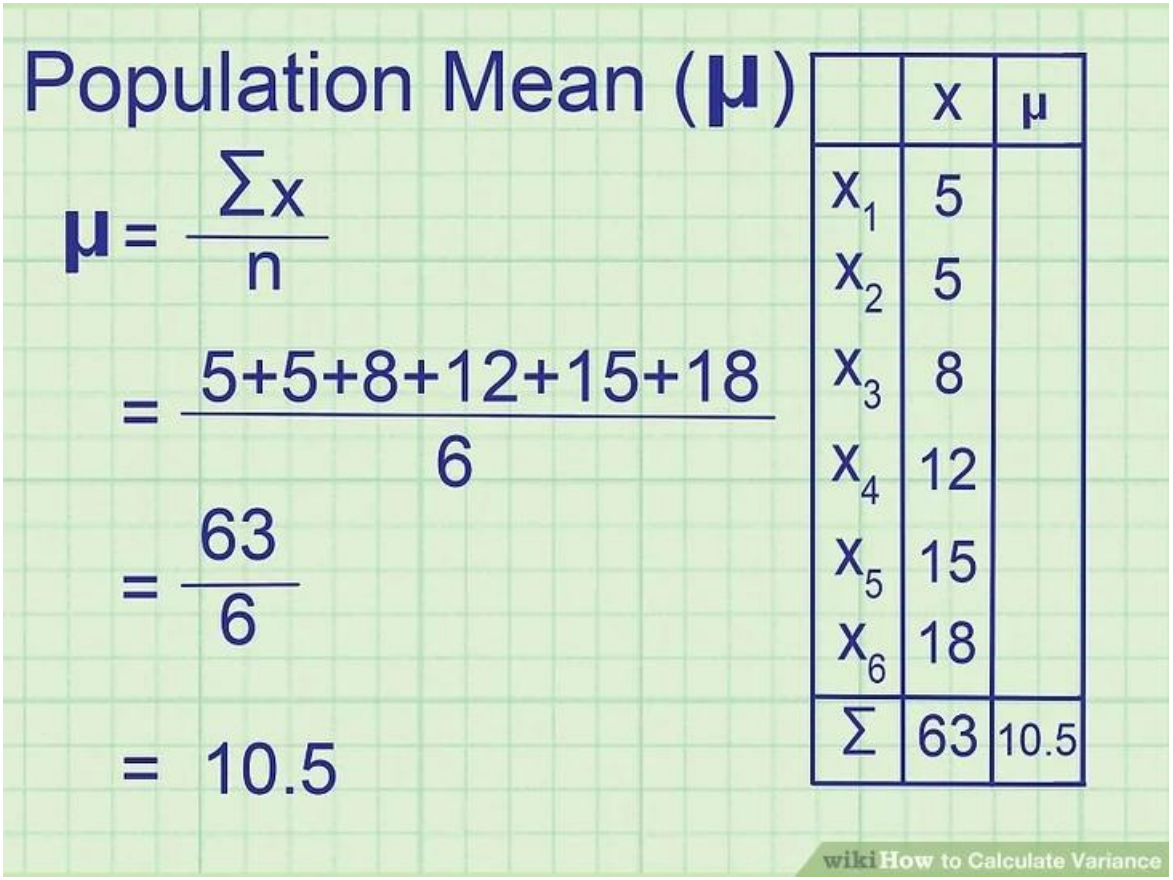


Calculate Population Variance



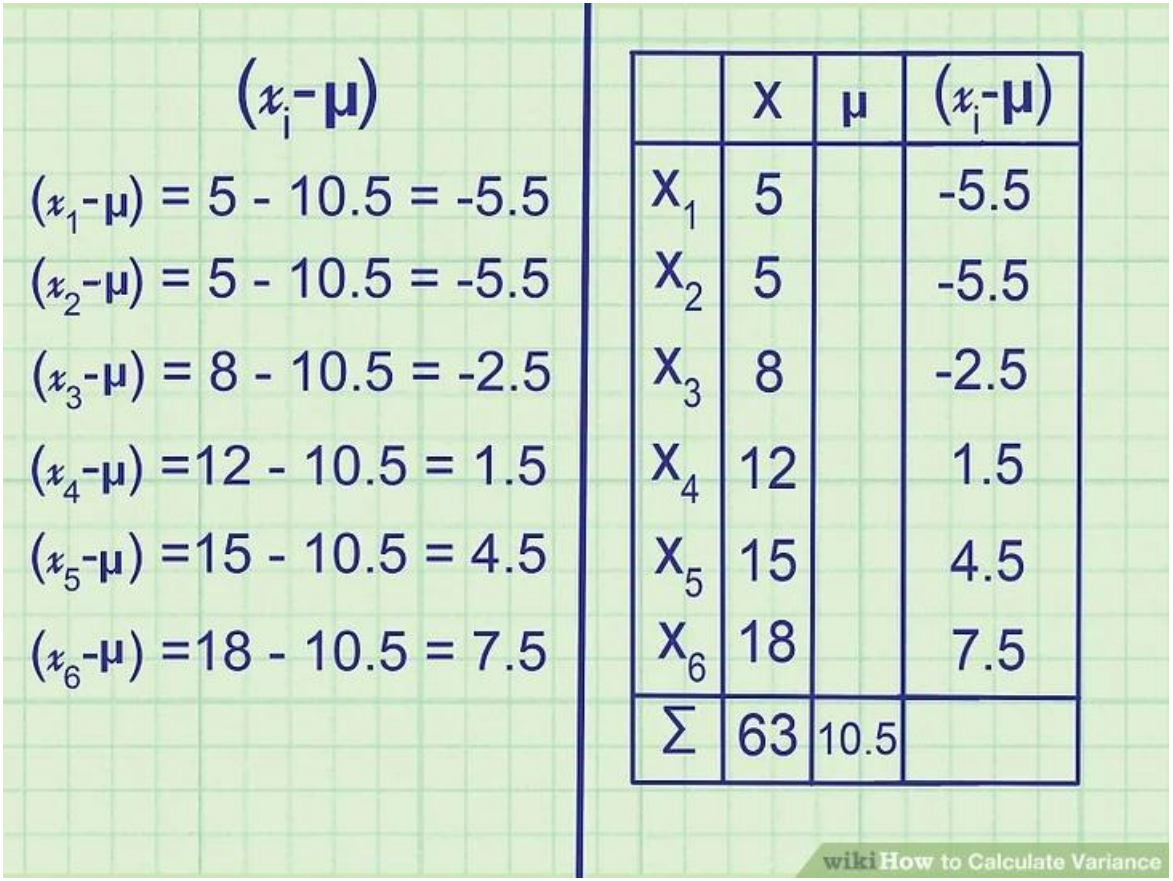
Calculate Population Variance

1. Find the mean of the Population data set. $\mu = \frac{\sum X}{N}$



Calculate Population Variance

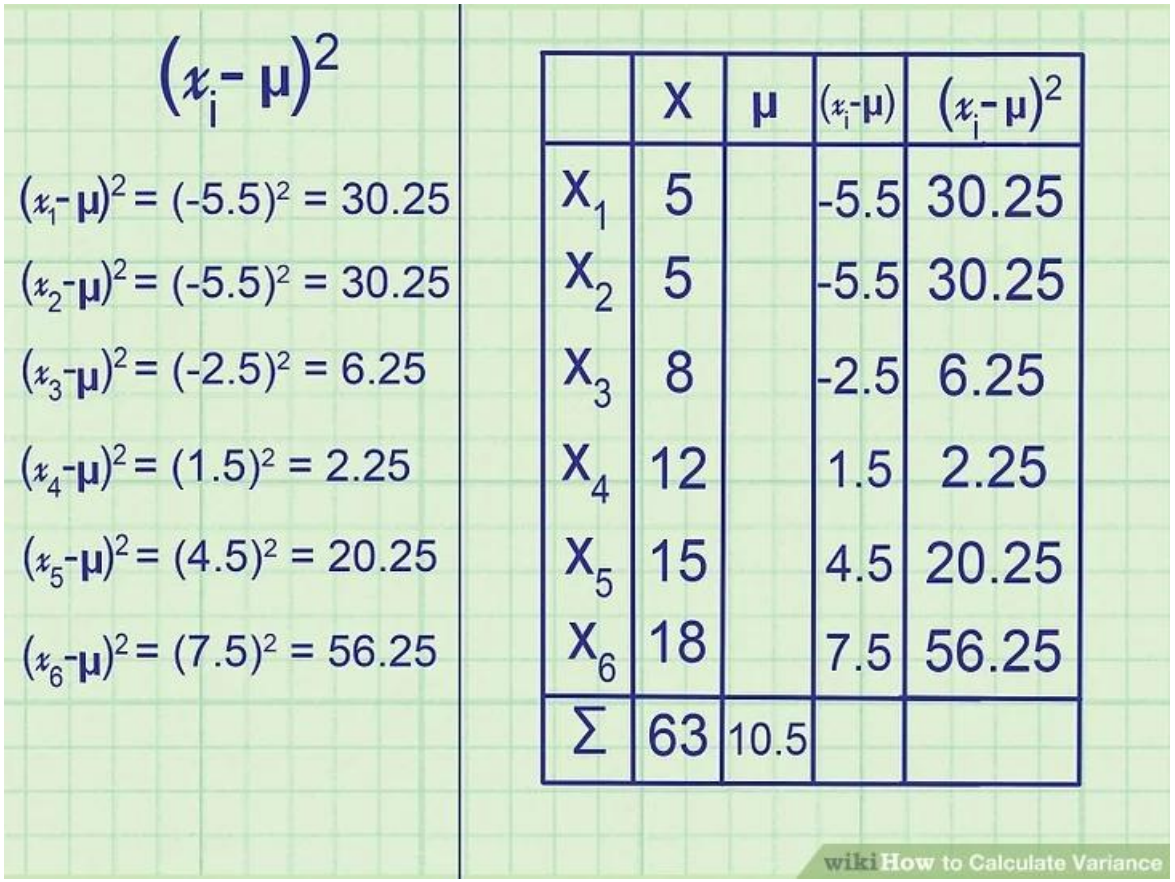
2. Find deviation of each entry. $(X - \mu)$



	X	μ	$(x_i-\mu)$
X_1	5		-5.5
X_2	5		-5.5
X_3	8		-2.5
X_4	12		1.5
X_5	15		4.5
X_6	18		7.5
Σ	63	10.5	

Calculate Population Variance

3. Square each deviation. $(X - \mu)^2$



	X	μ	$(x_i-\mu)$	$(x_i-\mu)^2$
X_1	5		-5.5	30.25
X_2	5		-5.5	30.25
X_3	8		-2.5	6.25
X_4	12		1.5	2.25
X_5	15		4.5	20.25
X_6	18		7.5	56.25
Σ	63	10.5		

Calculate Population Variance

4. Add to get the sum of the deviations squared. $\sum(X-\mu)^2$

	X	μ	$(x_i-\mu)$	$(x_i-\mu)^2$
X_1	5		-5.5	30.25
X_2	5		-5.5	30.25
X_3	8		-2.5	6.25
X_4	12		1.5	2.25
X_5	15		4.5	20.25
X_6	18		7.5	56.25
Σ	63	10.5		145.5

$(x_i-\mu)^2 = 30.25+30.25+6.25+2.25+20.25+56.25$

$(x_i-\mu)^2 = 145.5$

wikiHow to Calculate Variance

Calculate Population Variance

5. Divide by N to get the **Population variance**. $\sigma^2 = \frac{\sum(X-\mu)^2}{N}$

	x	μ	$(x_i-\mu)$	$(x_i-\mu)^2$
x_1	5		-5.5	30.25
x_2	5		-5.5	30.25
x_3	8		-2.5	6.25
x_4	12		1.5	2.25
x_5	15		4.5	20.25
x_6	18		7.5	56.25
Σ	63	10.5		145.5

$$\sigma^2 = \frac{145.5}{6}$$

$$= 24.25$$

wikiHow to Calculate Variance

Calculate Population Variance

6. Find the square root to get the **Population standard deviation**. $\sigma = \sqrt{\sigma^2}$

	x	μ	$(x_i-\mu)$	$(x_i-\mu)^2$
x_1	5		-5.5	30.25
x_2	5		-5.5	30.25
x_3	8		-2.5	6.25
x_4	12		1.5	2.25
x_5	15		4.5	20.25
x_6	18		7.5	56.25
Σ	63	10.5		145.5

$$\sigma = \sqrt{24.25}$$

$$\sigma^2 = \frac{145.5}{6}$$

$$= 24.25$$

$$= 4.92$$

wikiHow to Calculate Variance

Standard Error

Standard Error (SE)

- **What it is:** Measures how "reliable" your sample average is.
- **Think:** If you take the same size sample many times, how much would the averages jump around?
- **Tip:** Bigger sample → smaller SE → average is more reliable.

Standard Error

$$SE = \frac{s}{\sqrt{n}}$$

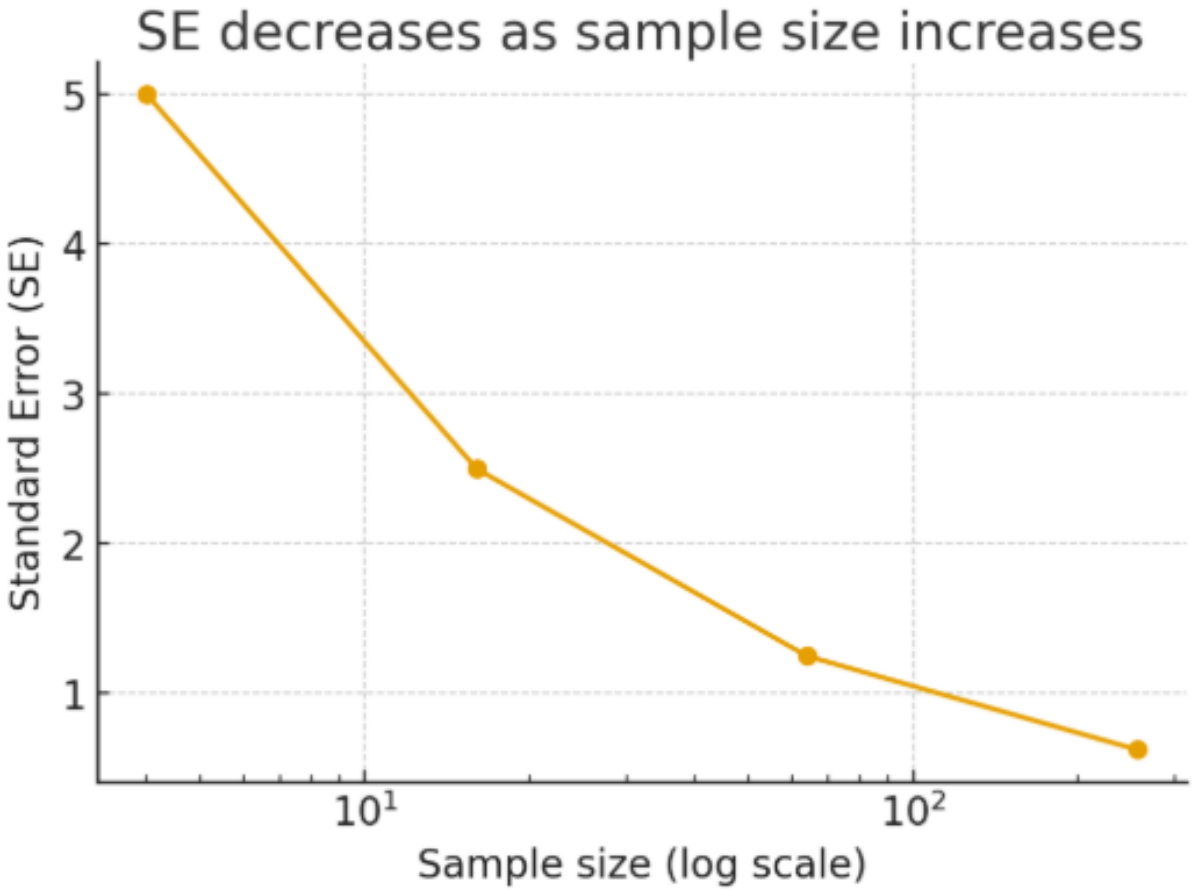
Where:

s= sample standard deviation

n= sample size

Key idea: Larger samples → smaller SE → more precise estimate of the population mean.

Standard Error



Standard Error

Assume the population standard deviation $\sigma = 10$. Compute SE for several sample sizes:

1. $n = 4$

$$\text{SE} = \frac{10}{\sqrt{4}} = 5$$

2. $n = 16$

$$\text{SE} = \frac{10}{\sqrt{16}} = 2.5$$

3. $n = 64$

$$\text{SE} = \frac{10}{\sqrt{64}} = 1.25$$

4. $n = 256$

$$\text{SE} = \frac{10}{\sqrt{256}} = 0.625$$

$$SE = \frac{s}{\sqrt{n}}$$

z-Scores as a Standardized Distribution

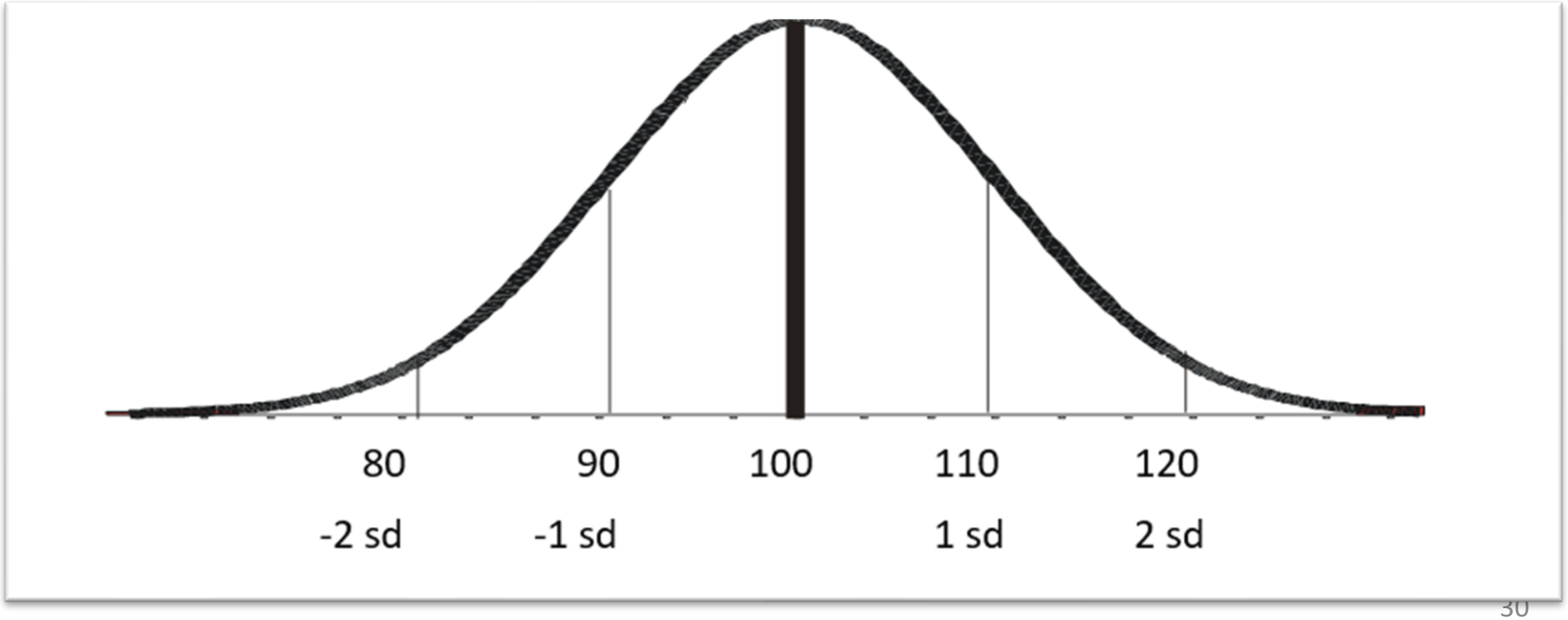
z-Scores as a Standardized Distribution

z-scores

- z-scores state the position of a raw score in relation to the mean.

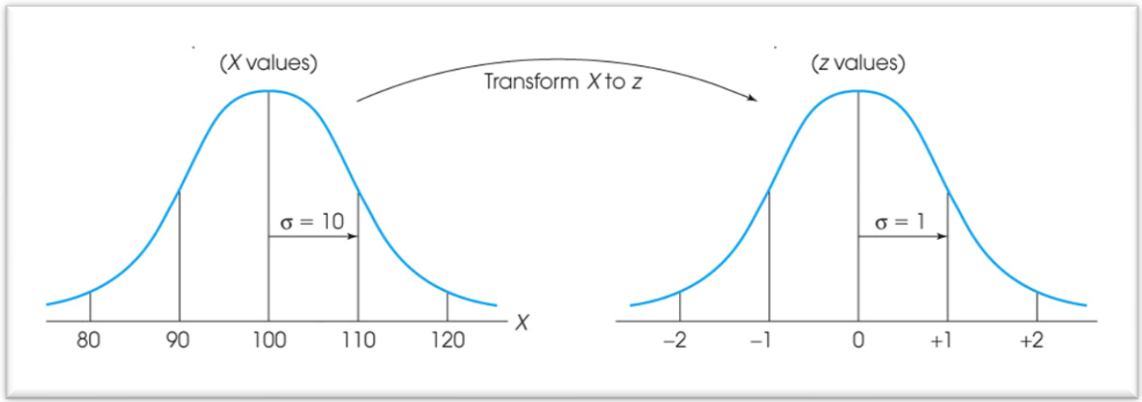
OR

- z-scores is the measure of an observation's (raw score) distance from the mean.



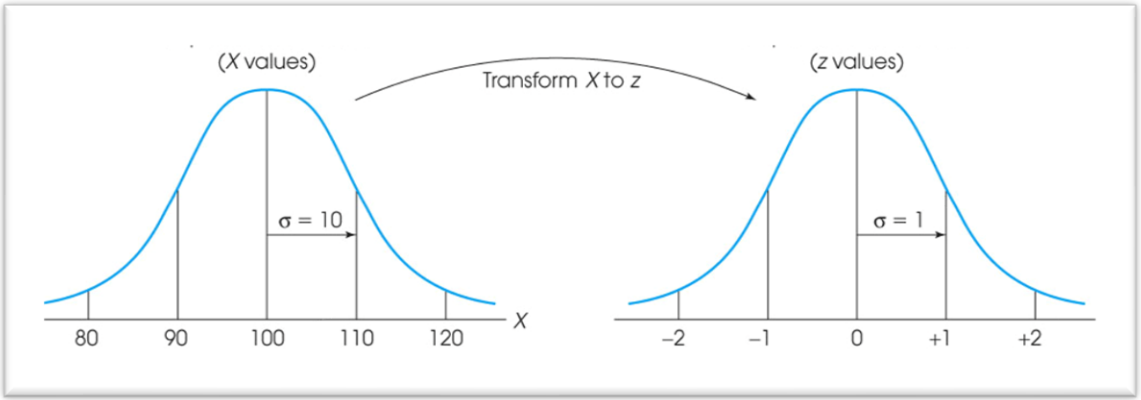
z-Scores and Location

- The raw score is transformed into a z-score, the z-score tells exactly where the score is located relative to all the other scores in the distribution.
- Z-scores will always have a mean of "0" (Zero).
- Z-scores will always have a standard deviation (σ) of "One" (1).
- The process of changing an X value into a z-score involves creating a signed number (+ or –).
- The sign of the z-score (+ or –) identifies whether the X value is located above the mean (positive) or below the mean (negative).



z-Scores and Location

- If a z-score is zero, it's on the mean.
- If a z-score is positive (+), it's above the mean.
- If a z-score is negative (-), it's below the mean.
- If a z-score is 1, it's 1 SD above the mean.
- If a z-score is 2, it's 2 SDs above the mean.
- If a z-score is -1, it's 1 SD below the mean.
- If a z-score is -2, it's 2 SDs below the mean.



z-Scores and Location

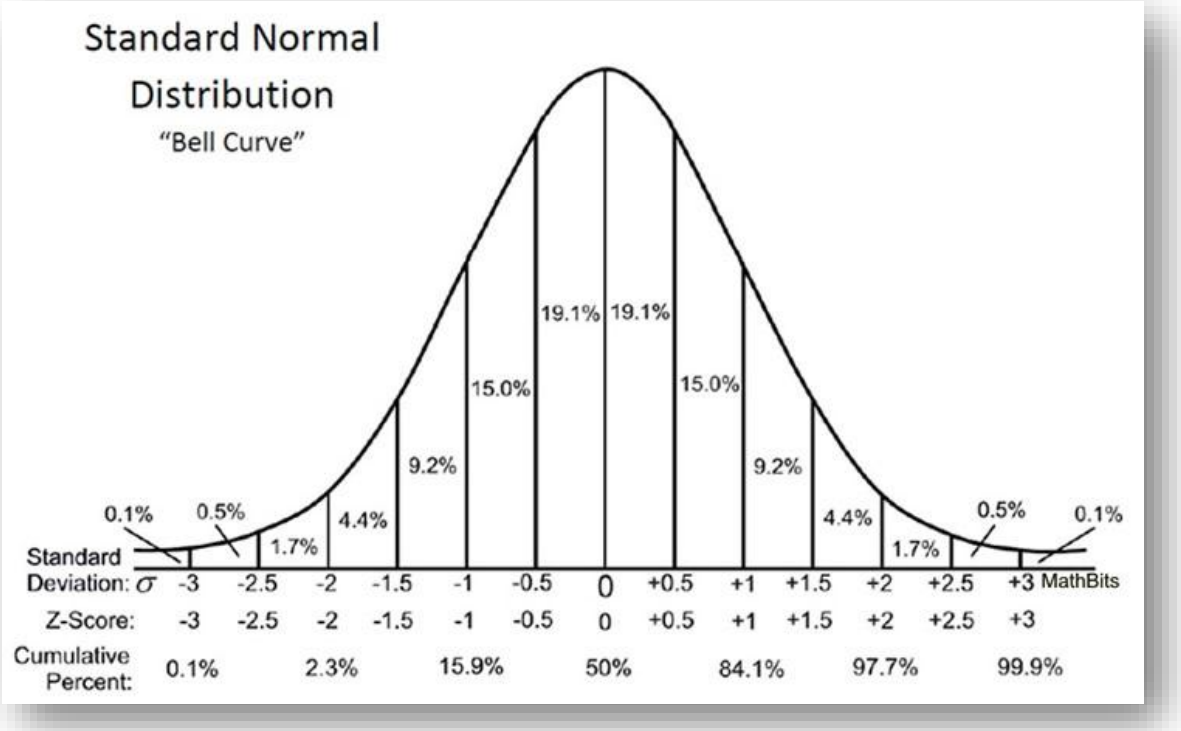
- If Z-scores less than -3 or greater than +3 are considered exceptional or unusual, and may indicate numerous interpretations:
- **Possible Interpretations**

1. **Outliers**
   - Data points that are **significantly different** from the rest of the dataset.
   - May represent natural variability or rare events.
2. **Errors**
   - Could result from **measurement inaccuracies**, **data entry mistakes**, or **instrumental faults**.
3. **Anomalies**
   - Indicate **unusual patterns or trends** that may require further investigation or validation.

z-Scores and Location

Z-Score Outlier Thresholds

- **Z < -3:**
  - ✓ Extremely low values
  - ✓ 0.1% of data points
- **Z > +3:**
  - ✓ Extremely high values
  - ✓ 0.1% of data points



z-Scores as a Standardized Distribution

$$z = \frac{Raw\ score - Mean}{Standard\ Deviation}$$

For a Population:

$$z = \frac{X - \mu}{\sigma}$$

For a Sample:

$$z = \frac{X - \bar{X}}{s}$$

Calculating z scores (***For a Population***)

The Z-score Formula

$$z = \frac{X - \mu}{\sigma}$$

For a distribution with a $\mu = 85$ and $\sigma = 5$

What is the z-score for X = 85? ---------CASE-1

$$z = \frac{85 - 85}{5} = \frac{0}{5} = 0$$

What is the z-score for X = 90? ---------CASE-2

$$z = \frac{90 - 85}{5} = \frac{5}{5} = 1$$

What is the z-score for X = 77? ---------CASE-3

$$z = \frac{77 - 85}{5} = \frac{-8}{5} = -1.6$$

Calculating z scores (***For a Population***)

- **Case-1**
- The z-score value is zero. So, it's on the mean.
- **Case-2**
- The z-score is positive (+). So, it's above the mean.
- The z-score value is 1. So, it's 1 SD above the mean.
- **Case-3**
- The z-score is negative (-). So, it's below the mean.
- The z-score value is -1.6. So, it's 1.6 SD below the mean.

Calculating z scores (***For a Sample***)

- **What is the z score for your test:** raw score = 80; mean = 75, *S* = 5?

$$z = \frac{X - \overline{X}}{S} \qquad z = \frac{80 - 75}{5} = 1$$

- **What is the z score of your friend's test:** raw score = 80; mean = 75, *S* = 10

$$z = \frac{X - \overline{X}}{S} \qquad z = \frac{80 - 75}{10} = .5$$

- **Who do you think did better on their test? Why do you think this?**

Calculating z scores (***For a Sample***)

- **Who do you think did better on their test? Why do you think this?**

Comparison of Z-Scores

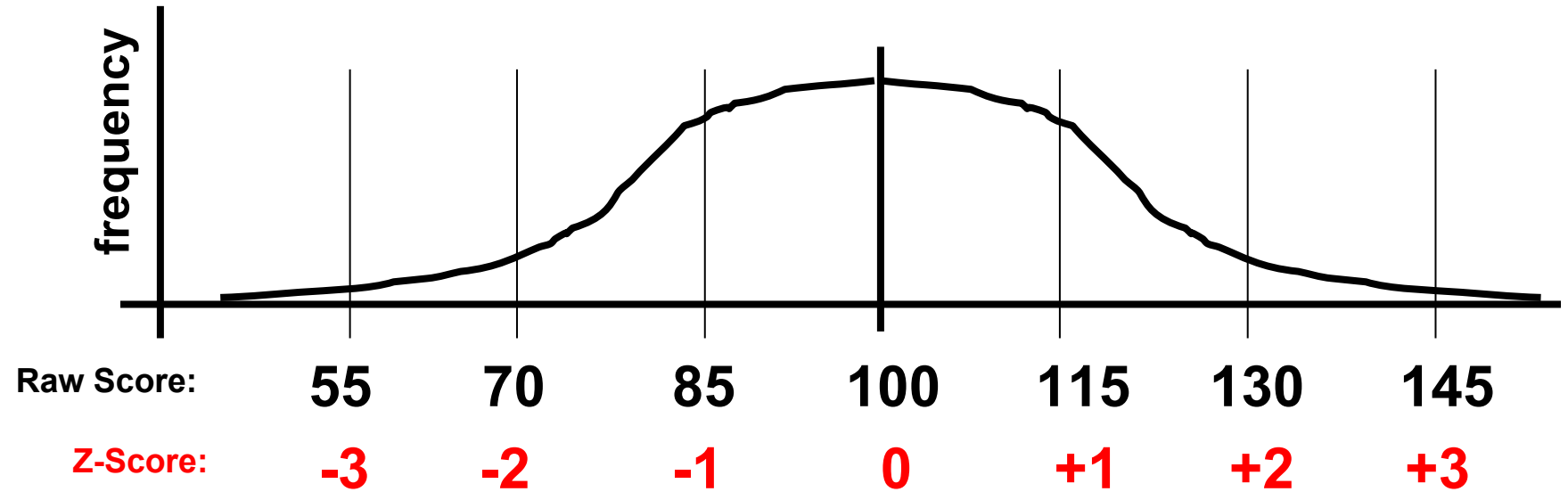
Person	Score	Mean	Standard Deviation	Z-Score
You	80	75	5	+1.0
Friend	80	75	10	+0.5

- **Comparison:**
  - ✓ Your Z-score: 1
  - ✓ Friend's Z-score: 0.5
- **Who did better?**
  - ✓ You did better on your test.
- **Why?**
  - ✓ Your Z-score of 1 indicates you scored 1 standard deviation above the mean, while your friend's Z-score of 0.5 indicates he scored only 0.5 standard deviations above the mean.
  - ✓ Therefore, your performance is relatively better.
- **Conclusion:**

✓ Your performance is **relatively stronger**, because your score was **further above the average** of your group compared to your friend's score in his group.

Transforming X to z-score

- **Z-scores** transform raw scores into standardized values that have:
- ✓ **Mean = 0**
- ✓ **Standard Deviation = 1**
- This allows comparison across different datasets or tests, regardless of their original scales.



Transforming X to z-score

$$z = \frac{x - \mu}{\sigma}$$

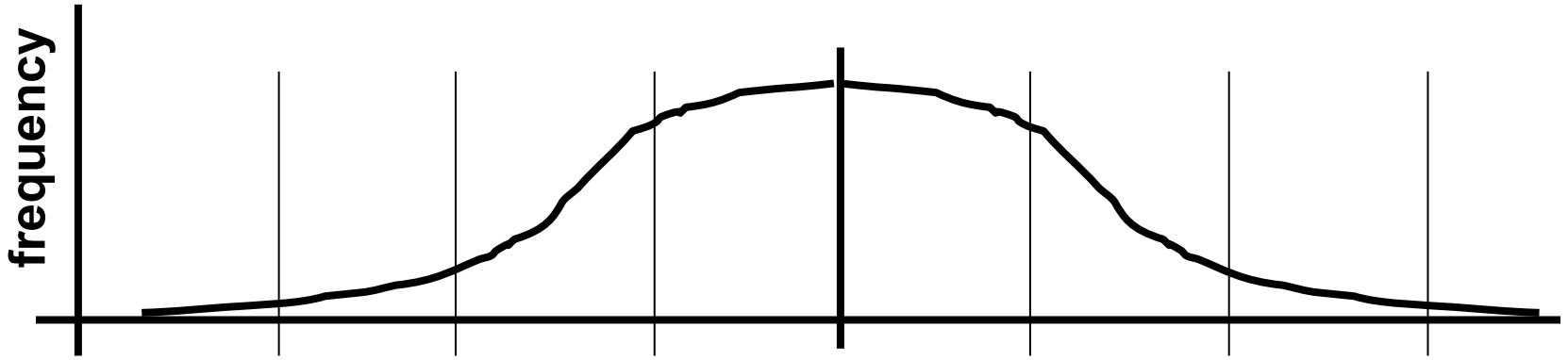
For raw scores: mean = 100, SD = 15

- **Z-score for 100:** (100-100) / 15 = 0,
- **Z-score for 85:** (85-100) / 15 = -1,
- **Z-score for 70:** (70-100) / 15 = -2,
- **Z-score for 55:** (55-100) / 15 = -3,

Z-score for 115: (115-100) / 15 = 1

Z-score for 130: (130-100) / 15 = 2

Z-score for 145: (145-100) / 15 = 3



Raw Score:	55	70	85	100	115	130	145
Z-Score:	-3	-2	-1	0	+1	+2	+3

Transforming X to z-score

$$z = \frac{x - \mu}{\sigma}$$

Example

Given:

- **Mean = 100**
- **Standard Deviation = 15**

Raw Score (X)	Formula	Z-Score	Interpretation
145	(145 – 100) / 15	**+3**	Extremely high (3 SD above mean)
130	(130 – 100) / 15	**+2**	Very high (2 SD above mean)
115	(115 – 100) / 15	**+1**	Above average (1 SD above mean)
100	(100 – 100) / 15	**0**	Average (at the mean)
85	(85 – 100) / 15	**–1**	Below average (1 SD below mean)
70	(70 – 100) / 15	**–2**	Low (2 SD below mean)
55	(55 – 100) / 15	**–3**	Extremely low (3 SD below mean)

Example

$$z = \frac{x - \mu}{\sigma}$$

- **Here are eight test scores from a Stats class: 35, 59, 70, 73, 75, 81, 84, 86.**
- **Mean = 70.4**
- **Standard deviation = 16.7**

a) Find the z-score of the data point 35.

z = -2.119

$$z = \frac{35 - 70.4}{16.7}$$

$$z = \frac{-35.4}{16.7} = -2.119$$

b) Find the z-score of the data point 73.

z = 0.1557

$$z = \frac{73 - 70.4}{16.7}$$

$$z = \frac{2.6}{16.7} = 0.1557$$

Application of z-scores (Why use z-scores?)

- **Test A:** Noura scores 78, Mean score = 70, SD = 8.
- **Test B:** Noura scores 78, Mean score = 66, SD = 6
- Did Noura do better or worse on **Test B**?

Test A:
Score (X) = 78
Mean (μ) = 70
Standard Deviation (σ) = 8
$Z = (X - \mu) / \sigma$
$= (78 - 70) / 8$
$= 8 / 8$
$= 1$

Test B:
Score (X) = 78
Mean (μ) = 66
Standard Deviation (σ) = 6
$Z = (X - \mu) / \sigma$
$= (78 - 66) / 6$
$= 12 / 6$
$= 2$

Comparison:

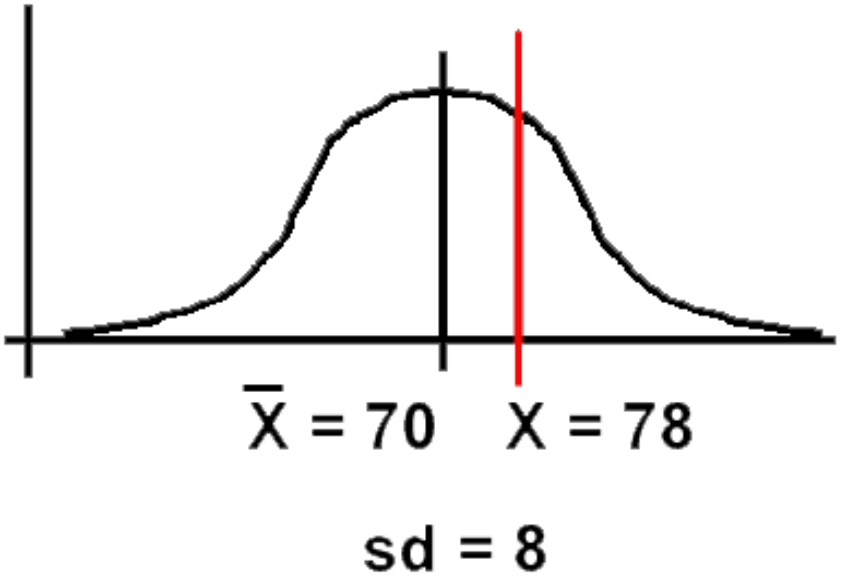
- ✓ **Test A:** Z = 1 (1 standard deviation above mean)
- ✓ **Test B:** Z = 2 (2 standard deviations above mean)

Conclusion:

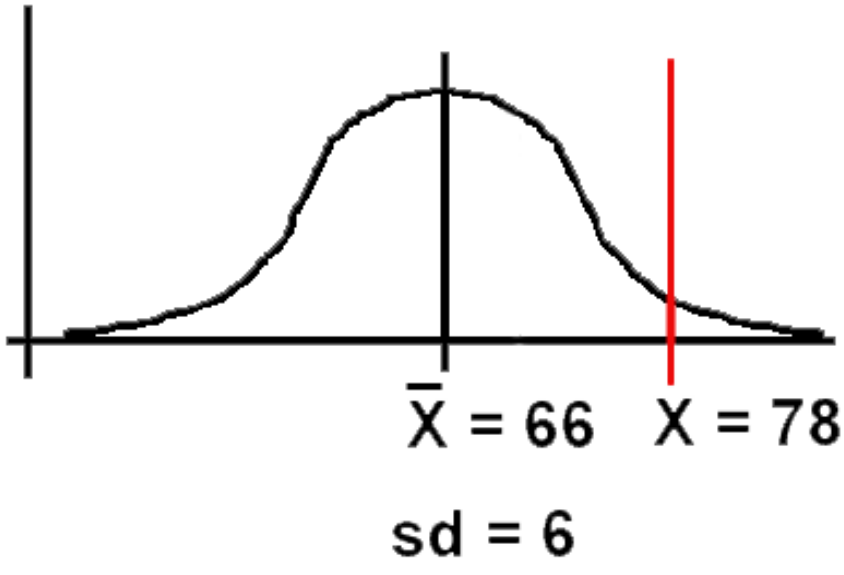
Noura did relatively better on **Test B**.

- ✓ **Test A:** z-score, z = 1.00
- ✓ **Test B:** z-score , z = 2.00

Conclusion: Noura did much better on Test B.



Test A

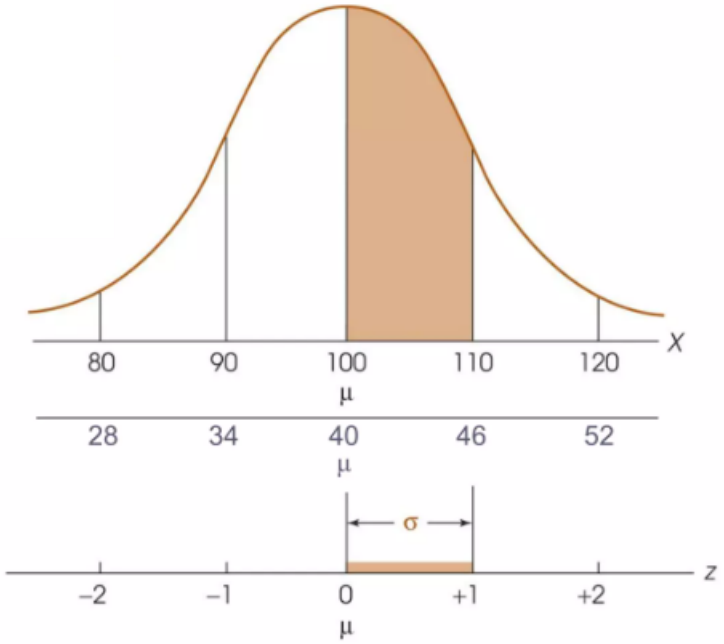


Test B

Advantage of Standardized Distributions

- Two (or more) different distributions can be made the same and compared to each other
- • Distribution A: $\mu = 100$, $\sigma = 10$
- • Distribution B: $\mu = 40$, $\sigma = 6$
- When these distributions are transformed to z-scores,

both will have a $\mu = 0$ and $\sigma = 1$



Z-Score outlier calculation

Z-Score outlier calculation

Definition:

- An outlier is a data point that significantly deviates from the majority of the data, exceeding conventional limits or patterns.

Z-Score outlier calculation

Step 1:

- Calculate Mean (μ)

$$\mu = (\Sigma X) / n$$

Step 2:

- Calculate Standard Deviation (σ)

$$\sigma = \sqrt{\frac{\Sigma(x-\mu)^2}{n-1}}$$

Step 3:

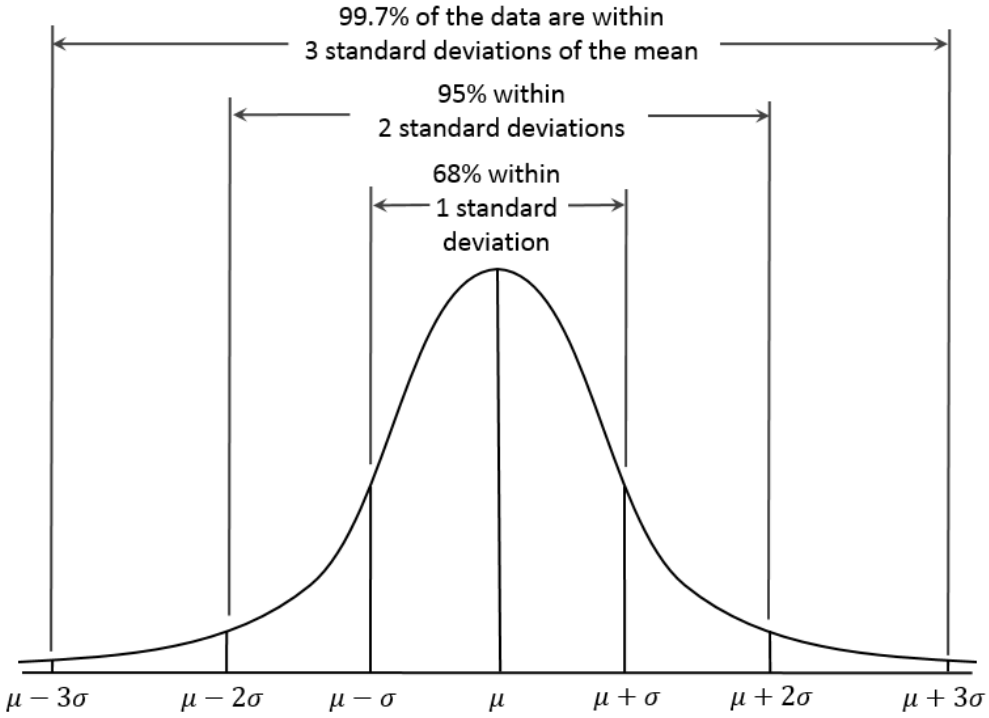
Calculate Z-Score

$$Z = (X - \mu) / \sigma$$

Z-Score outlier calculation

Outlier Thresholds:

- $Z < -3$ or $Z > 3$: Extreme outliers (99.7% confidence)
- $Z < -2$ or $Z > 2$: Moderate outliers (95% confidence)
- $Z < -1$ or $Z > 1$: Mild outliers (68% confidence)



Z-Score outlier calculation

📈 Outlier Thresholds Based on Z-Scores

Z-Score Range	Classification	Approx. % of Data Within Range	Confidence Level	Interpretation
Z < −3 or Z > +3	**Extreme Outliers**	99.7%	Very High (≈ 99.7%)	Data points are extremely rare; likely errors or exceptional values.
Z < −2 or Z > +2	**Moderate Outliers**	95%	High (≈ 95%)	Unusual but possible values; may need verification.
Z < −1 or Z > +1	**Mild Outliers**	68%	Moderate (≈ 68%)	Slightly above or below average; usually within normal variation.

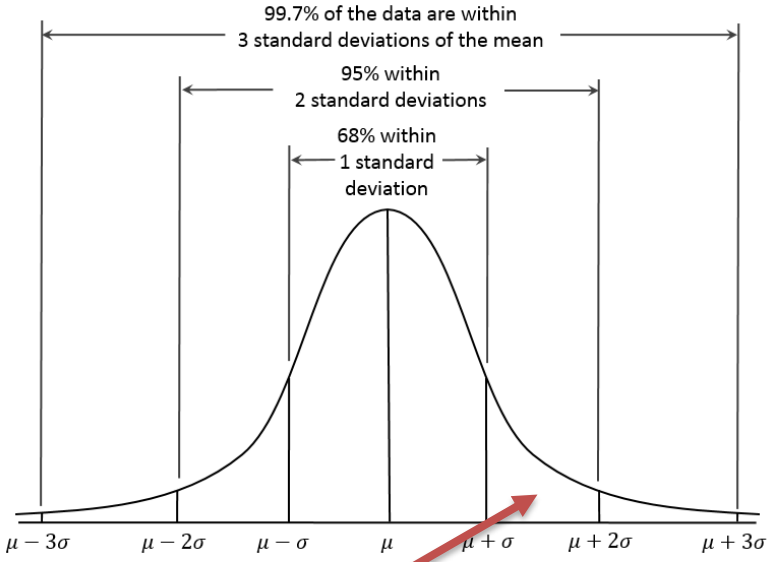
Z-Score outlier calculation

Example:

- Data: 1, 3, 5, 5, 5, 7, 9
- $\mu = 5$
- $\sigma = 2.58$
- Calculate Z-Score for X = 9:

$$Z = (X - \mu) / \sigma$$

$$Z = (9 - 5) / 2.58 \approx 1.55$$

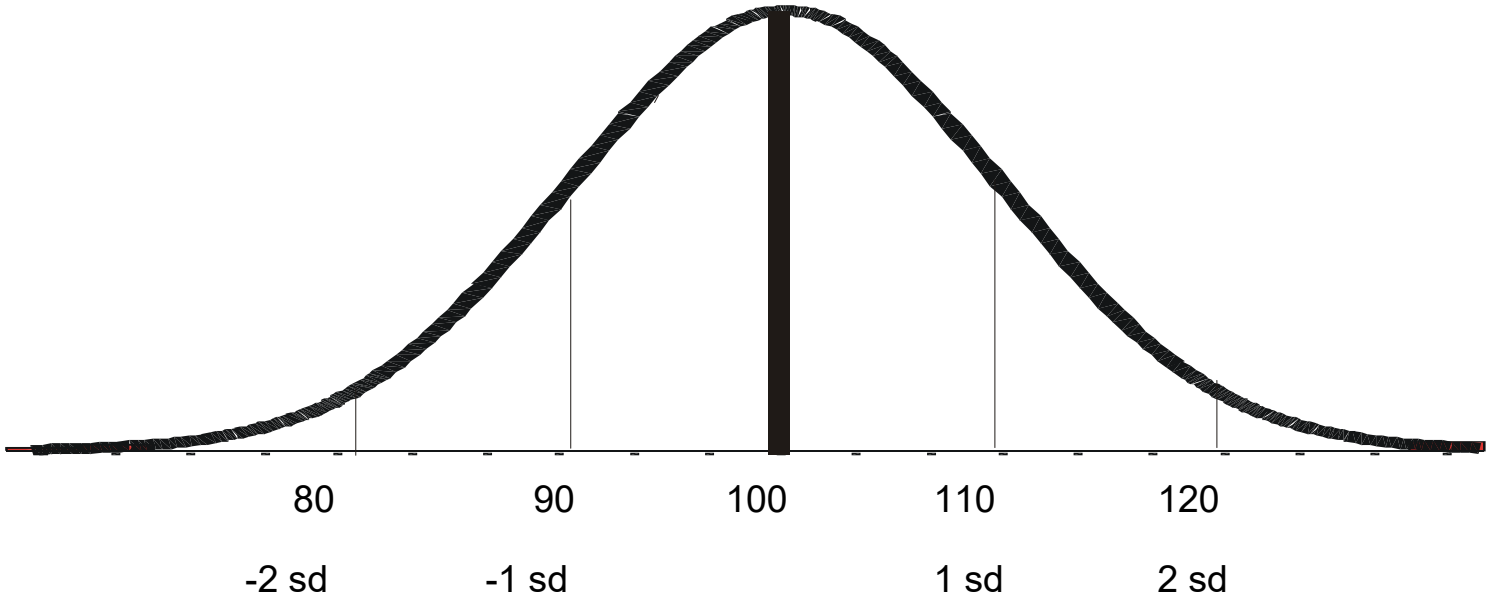


- Interpretation:-
  - ✓ Within 1.55 standard deviations from the mean

PRACTICE

Using standard deviation units to describe individual scores

Here is a distribution with a mean of 100 and standard deviation of 10:



What score is one sd below the mean?

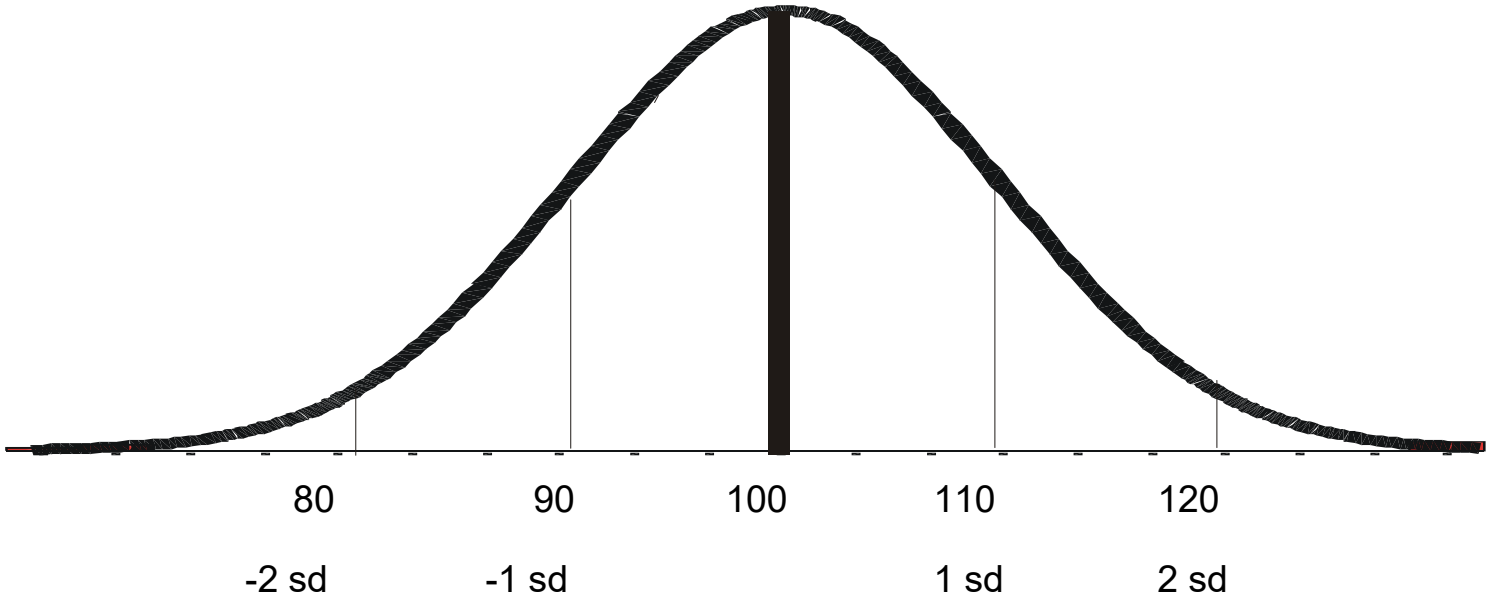
90

What score is two sd above the mean?

120

Using standard deviation units to describe individual scores

Here is a distribution with a mean of 100 and standard deviation of 10:



How many standard deviations below the mean is a score of 90?

How many standard deviations above the mean is a score of 120?

1
2

Fun facts about z scores

- The mean of a distribution has a z score of ____?
- Positive z scores represent raw scores that are _____ (above or below) the mean?
- Negative z scores represent raw scores that are _____ (above or below) the mean?

zero

above

below

THANK YOU